

Project 2 — An Agent That Plays a Classic Game

Fundamentos de Aprendizagem Automática

Mário Antunes

2026

1. Goal

Build an **agent that plays a traditional arcade game** and learns, or is designed, to play it well. Choose one game, for example **Frogger** or **Space Invaders**, or a game of similar size agreed with your instructor (Pong, Breakout, Pac-Man, Asteroids, ...).

The project is **open on purpose**: the technique is yours to choose and to defend. Several routes are valid, and you may combine them:

- reinforcement learning (Class 13): value-based or policy-based, on pixels or on a compact state;
- search and planning, with a model of the game or a learned model of its dynamics;
- evolutionary or other blind optimization of a policy (Class 02);
- supervised or imitation learning from recorded play;
- a hand-designed controller or heuristics, as a baseline or as part of the agent.

2. What We Expect

- **A working agent**, with a clear definition of the game, the observations, the actions and the reward.
- **Baselines**: at least a random agent and a simple scripted or heuristic one.
- **An evaluation you define and justify**: score, survival, success rate, learning curves, many runs with different seeds, the spread and not only the mean.

- **An analysis:** what the agent learned or exploits, where and why it fails, and how the design choices (representation, reward, algorithm, hyper-parameters) change the result.

3. Deliverables

- The **code**, reproducible with the course environment.
- A **README.md** with the same structure as Project 1: identification, the game and the agent (choices and justification), methodology, evaluation and conclusion, with plots and diagrams.
- A **short live demonstration** of the agent, with the class presentation (Class 14).

4. To Be Defined at the Release

These points are not closed yet and will be fixed when the project is released (Class 09):

- the game environment and the library that provides it, and the compute limits;
- the exact evaluation protocol and the baseline scores;
- the assessment criteria, the group size and the format of the presentation.

5. Dates

- **Release:** Class 09 (12/13 November 2026)
- **Clinic:** Class 13 (10/11 December 2026)
- **Submission deadline and presentations:** Class 14 (17/18 December 2026)
- **Weight:** 25% of the final grade